OPENSOURCESDRLAB DOCUMENTATION

FMC-ADRV9009 Quick Start Guide

Product	FMC-ADRV9009
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1. Overview

The FMC-ADRV9009 is an FPGA Mezzanine Card (FMC, HPC connector) built around the Analog Devices ADRV9009 RF transceiver and the AD9528 clock generator. It brings the transceiver's two transmit, two receive and two observation-receive channels out to SMA connectors and connects to the carrier FPGA through JESD204B.

This guide shows how to bring the board up on a Xilinx ZC706 carrier with the ADRV9009 Transceiver Evaluation Software, and where to find the bare-metal reference designs in this repository.

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2. What you need

Item	Notes
FMC-ADRV9009 board	
Carrier board	Xilinx ZC706 (used in this guide), or the XC7Z100 "7100" carrier
microSD card	16 GB or larger
Ethernet cable and USB-UART cable	for the ZC706 network and serial console
Windows PC	runs the evaluation software and the SD-card writer
SD evaluation image	`ADRV9009-eval-SD-image-20161130.img.xz` from the Releases page
Boot files and test configuration	`FMC-ADRV9009_eval-boot-files.zip` from the Releases page
ADRV9009 Transceiver Evaluation Software	download from Analog Devices: https://www.analog.com/en/products/adrv9009.html
SD-card writing tool	any image writer, for example Win32 Disk Imager or balenaEtcher

3. Board connectors

Connector	Function
TX1, TX2	transmit outputs
RX1, RX2	receive inputs
ORX1, ORX2	observation-receiver inputs
REF_IN	external reference clock input
EXT_LO	external local-oscillator input
FMC (HPC)	JESD204B data, SPI control and power from the carrier

See `hardware/FMC-ADRV9009_Schematic.pdf` for the full schematic.

4. Bring-up with the evaluation software (ZC706)

1. Decompress `ADRV9009-eval-SD-image-20161130.img.xz` (for example with 7-Zip) and write the `.img` file to the microSD card with your SD-card writer.

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2. When writing has finished, open the card's **BOOT** partition and copy every file from the `zc706_update` folder of `FMC-ADRV9009_eval-boot-files.zip` into it, replacing the existing files.
3. Insert the card into the ZC706 and set the boot-mode switches to **SD card boot**.
4. Fit the FMC-ADRV9009 on the ZC706 **HPC** FMC connector, connect the Ethernet cable and the USB-UART cable, then power on.
5. On the serial console, give the board an IP address:

   ```
   ifconfig eth0 192.168.1.10
   ```

   Set your PC to an address in the same subnet, for example `192.168.1.100`.
6. Start the ADRV9009 Transceiver Evaluation Software, connect to `192.168.1.10`, and update the firmware when the software asks you to.
7. Load the test configuration `ADRV9009_Test_MAX.xml` (included in `FMC-ADRV9009_eval-boot-files.zip`).
8. Set the transmit waveform and check the received waveform.

> **Tip:** if the received spectrum shows many spurious signals, reduce the receiver gain.

5. Reference designs in this repository

Folder	Carrier	Contents
`vivado/ZC706/`	Xilinx ZC706 (XC7Z045)	Vivado 2021.1 project (PL) and bare-metal application source (PS)
`vivado/7100/`	XC7Z100 "7100" carrier	Vivado 2021.1 project (PL) and bare-metal application source (PS)

The repository holds the project sources. Prebuilt files for each carrier (bitstream, hardware platform `.xsa`, FSBL and application `.elf`) are attached to the Releases page.

To rebuild from source:

1. Open `FMC_ADRV9009_PL/FMC_ADRV9009_PL.xpr` in Vivado 2021.1, generate the bitstream and export the hardware (`.xsa`).
2. Download the two Talise firmware headers from Analog Devices into `FMC_ADRV9009_PS/src/firmware/`. They are not part of this repository; `src/firmware/README.md` gives the links (ADI no-OS, branch `2021_R1`, firmware 6.0.2) and the SHA-256 values to check.

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3. In Vitis 2021.1, create a platform from the `.xsa` (standalone, `ps7_cortexa9_0`), then an empty bare-metal C application, and import everything in `FMC_ADRV9009_PS/src` (including `lscript.ld`).

4. In the application's C/C++ build settings, add these include directories (ARM v7 gcc compiler > Directories):

   ```
   src/app
   src/devices
   src/devices/adi_hal
   src/devices/talise
   src/firmware
   ```

   The linker libraries are the Vitis defaults (`-Wl,--start-group,-lxil,-lgcc,-lc,--end-group`).

5. Build the application, then program the bitstream and run the `.elf` over JTAG, or create `BOOT.bin` with the FSBL, bitstream and application to boot from the SD card.

6. Support

- Product page: https://opensourcesdrlab.com/products/opensourcesdrlab-new-fmc-adrv9009-board-high-performance-board
- Email: support@opensourcesdrlab.com
- Forum: https://forum.opensourcesdrlab.com
- Discord: https://discord.gg/nnxY6DkmvT

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